Deep Generative Models
Introduction

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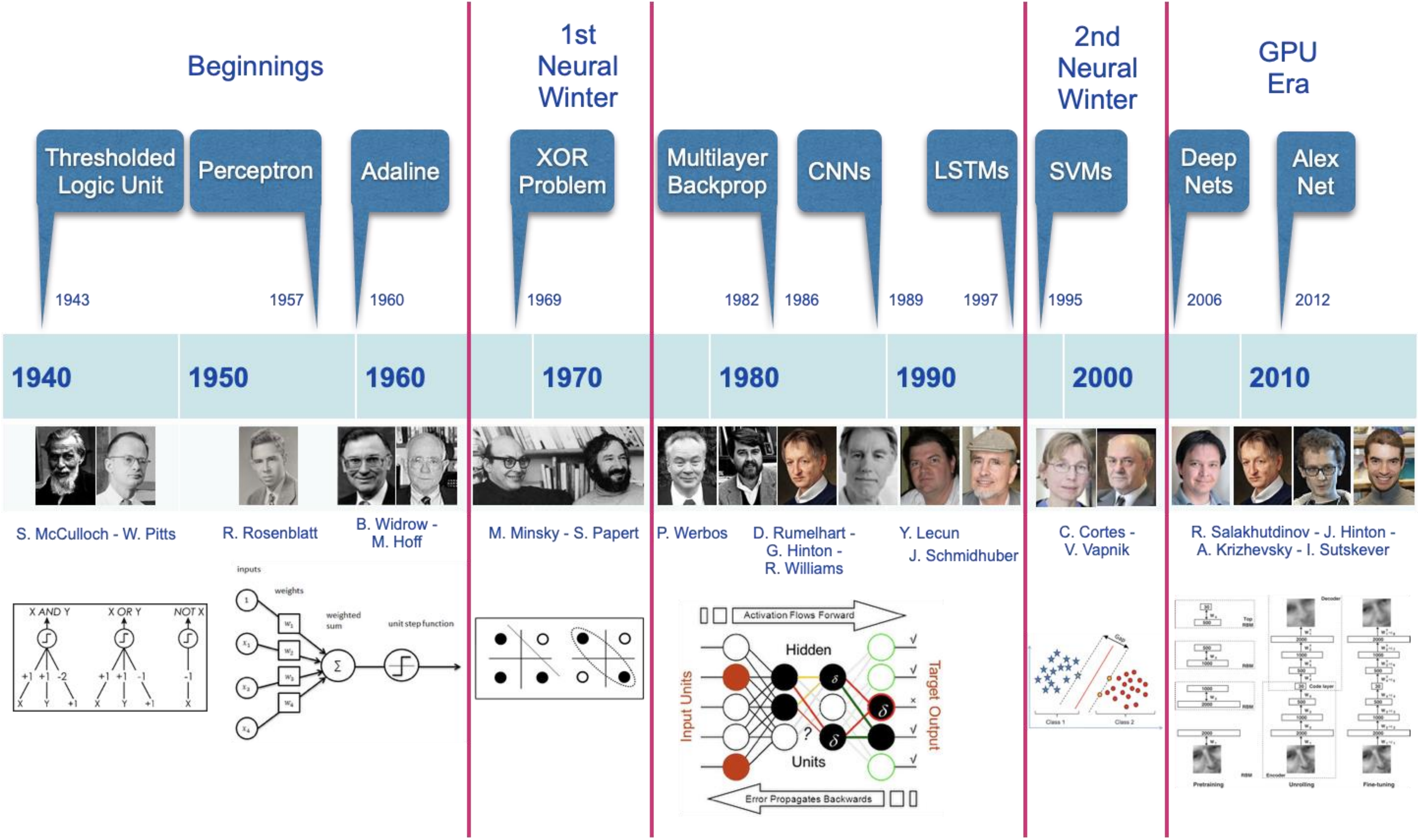
Outline

- History of AI
- History of Generative Models
- Advent of Deep Generative Models
- Applications of Deep Generative Models


Beginnings
1st Neural Winter
2nd Neural Winter
GPU Era
Thresholded Logic Unit
Perceptron
Adaline
XOR Problem
Multilayer Backprop
CNNs
LSTMs
SVMs
Deep Nets
Alex Net
1943
1957
1960
1969
1982
1986
1989
1997
1995
2006
2012
1940
1950
1960
1970
1980
1990
2000
2010
S. McCulloch - W. Pitts
R. Rosenblatt
B. Widrow - M. Hoff
M. Minsky - S. Papert
P. Werbos
D. Rumelhart - G. Hinton - R. Williams
Y. Lecun
J. Schmidhuber
C. Cortes - V. Vapnik
R. Salakhutdinov - J. Hinton - A. Krizhevsky - I. Sutskever
X AND Y
X OR Y
NOT X
inputs
weights
weighted sum
unit step function
Activation Flows Forward
Hidden
Units
Input Units
Target Output
Error Propagates Backwards
Pretraining
Unrolling
Fine-tuning

Generative Models, 1950–1980

	FOUNDATIONS	IMAGES	VIDEO	TEXT
1950	**Information theory** Shannon: prediction and entropy	**Statistical image coding** Pixels as random signals	—	**Statistical language models** Shannon: letter and word prediction
1957–58	**Perceptron** Rosenblatt	**Trainable visual patterns** Early neural vision	—	**Generative grammar** Chomsky, 1957
1962–66	**PCA / K–L models; HMMs** Latent vectors and sequences	**Linear latent image models** K–L image coding	**Probabilistic sequence roots** Hidden Markov models	**ELIZA** Weizenbaum, 1966
1970–74	**Spatial lattice models** Besag, 1974	**Statistical texture synthesis** Julesz-style statistics	**Autoregressive temporal models** Frames as stochastic sequences	**SHRDLU; probabilistic grammars** Structured generation
1975–79	**Latent-variable estimation** EM algorithm, 1977	**Model-based vision** Shape and texture primitives	**Motion as a state process** State-space foundations	**Rule-based text generation** Templates and grammars

Generative Models, 1980–2010

	FOUNDATIONS	IMAGES	VIDEO	TEXT
1982–86	**Energy models + latent codes** Hopfield; Boltzmann; autoencoders	**MRF texture models** Cross–Jain; Geman–Geman	**State-space motion models** Stochastic dynamics	**Neural distributed representations** PDP / backpropagation
1990–95	**Directed latent models** Helmholtz machine; wake–sleep	**Eigenfaces; deformable models** Turk–Pentland; active shapes	**Probabilistic temporal networks** HMMs and dynamic Bayes nets	**Statistical n-gram LMs** Corpus-trained language generation
1997–2000	**Sampling + learned latents** Probabilistic neural synthesis	**AAMs; texture synthesis** Cootes et al.; Efros–Leung	**Video Rewrite; Video Textures** Bregler et al.; Schödl et al.	**Recurrent neural language models** Elman-style sequence learning
2001–05	**Kernel and graphical models** Richer learned distributions	**Image quilting; analogies** Efros–Freeman; Hertzmann et al.	**Dynamic textures** Doretto et al., 2003	**Neural probabilistic LM; LDA** Bengio et al.; Blei et al., 2003
2006–09	**Deep belief nets + deep AEs** Hinton et al., 2006	**Deep Boltzmann machines** Salakhutdinov–Hinton, 2009	**Gated temporal models** Memisevic–Hinton; Taylor et al.	**Neural sequence generation** RNN language modeling revives

Generative Models, 2010–2020

	FOUNDATIONS	IMAGES	VIDEO	TEXT
2010–12	**Neural autoregressive models** NADE and deep density models	**Deep generative image models** DBMs and autoregression	**Recurrent video prediction** Learned temporal dynamics	**RNN language models** Mikolov et al., 2010
2013–14	**VAE + GAN** Kingma–Welling; Goodfellow et al.	**Neural latent image generation** VAEs and adversarial samples	**Learned video prediction** Frames to latent dynamics	**Sequence-to-sequence** Sutskever et al., 2014
2015–16	**Diffusion + autoregressive pixels** Sohl-Dickstein; PixelRNN/CNN	**DCGAN; pix2pix** Convolutional GANs	**Video GAN / TGAN** Vondrick et al.; Saito et al.	**Attention-based generation** Conditional sequence decoding
2017–18	**VQ-VAE + Transformer** Discrete latents; self-attention	**CycleGAN; StyleGAN** Translation and controllable style	**MoCoGAN; vid2vid** Motion/content + translation	**Transformer; GPT** Vaswani et al.; Radford et al.
2019	**Score-based generation** Song & Ermon	**StyleGAN2** High-fidelity face synthesis	**DVD-GAN** Large-scale video GANs	**GPT-2; BART; T5** Large pretrained generators

Generative Models, 2020–2026

	FOUNDATIONS	IMAGES	VIDEO	TEXT
2020	**DDPM + scaling laws** Ho et al.; generative scaling	**High-quality diffusion** DDPM; improved GANs	**Video transformers emerge** Discrete visual tokens	**GPT-3** Few-shot text generation at scale
2021	**Classifier-free guidance** Strong conditional sampling	**DALL·E; guided diffusion** Text-to-image scaling	**VideoGPT** VQ-VAE + Transformer	**Instruction tuning** Natural-language task control
2022	**Latent diffusion** Efficient high-resolution synthesis	**DALL·E 2; Imagen; Stable Diffusion** General text-to-image	**Video Diffusion; Imagen Video** Also Make-A-Video	**InstructGPT; ChatGPT; PaLM** Aligned conversational generation
2023	**Diffusion transformers** DiT-style scalable backbones	**SDXL; DALL·E 3** Better prompts and typography	**Gen-2; Emu; Stable Video Diffusion** Practical text/image-to-video	**GPT-4; Claude; Llama 2** Multimodal and open LLM era
2024	**Flow matching + multimodal tokens** Unified generative backbones	**FLUX; Stable Diffusion 3** Transformer / flow image models	**Sora; Veo; Movie Gen** Large video foundation models	**Long-context + reasoning models** Claude 3; Gemini 1.5; o1
2025–26	**Unified multimodal generation** Text, image, video and audio	**Native multimodal image models** Generation + editing + reasoning	**Veo 3/3.1; Sora 2; Genie 3** Audio, physics, interactive worlds	**Reasoning and agentic generators** Tools, memory, multimodal output

Stochastic Grammars

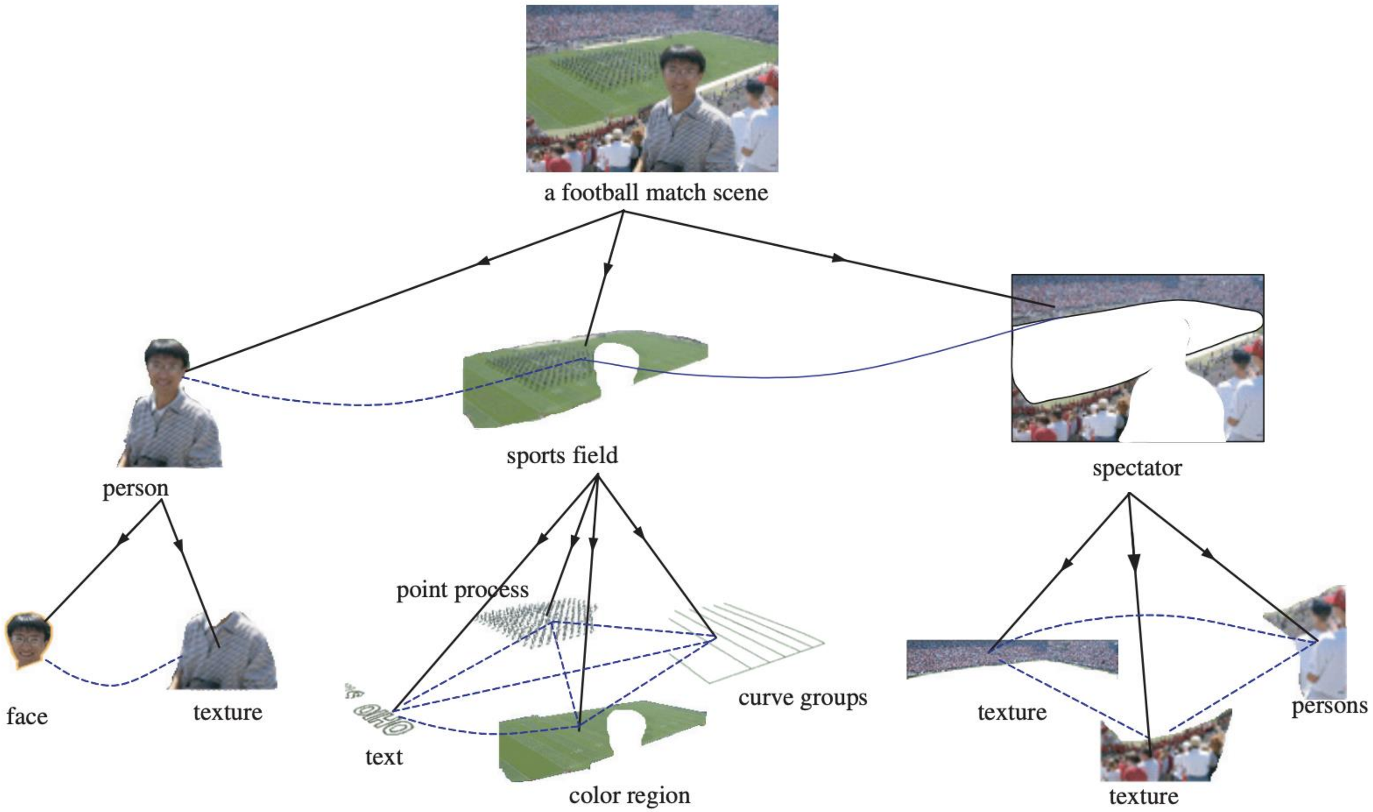


Fig. 1.1 Illustrating the task of image parsing. The parse graph includes a tree structured decomposition in vertical arrows and a number of spatial and functional relations in horizontal arrows. From [72].

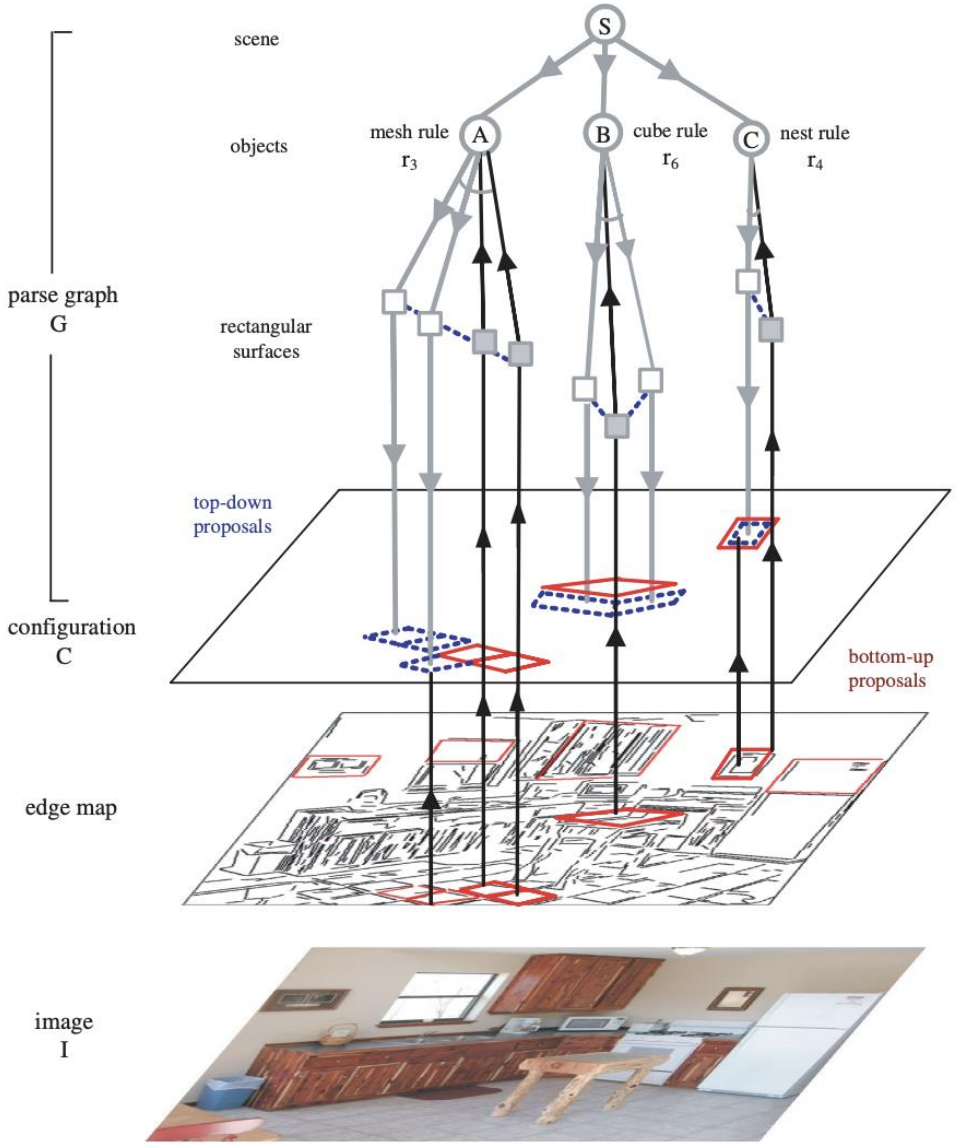


Fig. 1.2 Illustrating the recursive bottom-up/top-down computation processes in image parsing. The detection of rectangles (in red) instantiates some non-terminal nodes shown as upward arrows. They in turn activate graph grammar rules for grouping larger structures in nodes A, B, and C, respectively. These rules generate top-down prediction of rectangles (in blue). The predictions are validated from the image under the Bayesian posterior probability. Modified from [59].

Topic Modeling

what about topic modeling

Topic modeling is a significant area of generative models, particularly in text analysis and natural language processing. It was developed and popularized in the late 1990s and early 2000s. Here's how it fits into the history of generative models:

Topic Modeling Timeline (1950-2000)

1980s: Early Foundations

- **1980s: Latent Semantic Analysis (LSA)**. Although not strictly generative, LSA provides a foundation for topic modeling by analyzing the relationships between a set of documents and terms, reducing dimensions to uncover latent structures in the data.

1990s: Development and Formalization

- **1999: Latent Dirichlet Allocation (LDA)**. Introduced by Blei, Ng, and Jordan, LDA is a probabilistic model that represents documents as mixtures of topics, and topics as mixtures of words. This approach became a cornerstone of topic modeling and generative text models

2000s: Expansion and Application

- **2003: Topic Modeling with LDA**. The LDA model gains traction in various applications, from document classification to information retrieval, influencing subsequent research and tools.

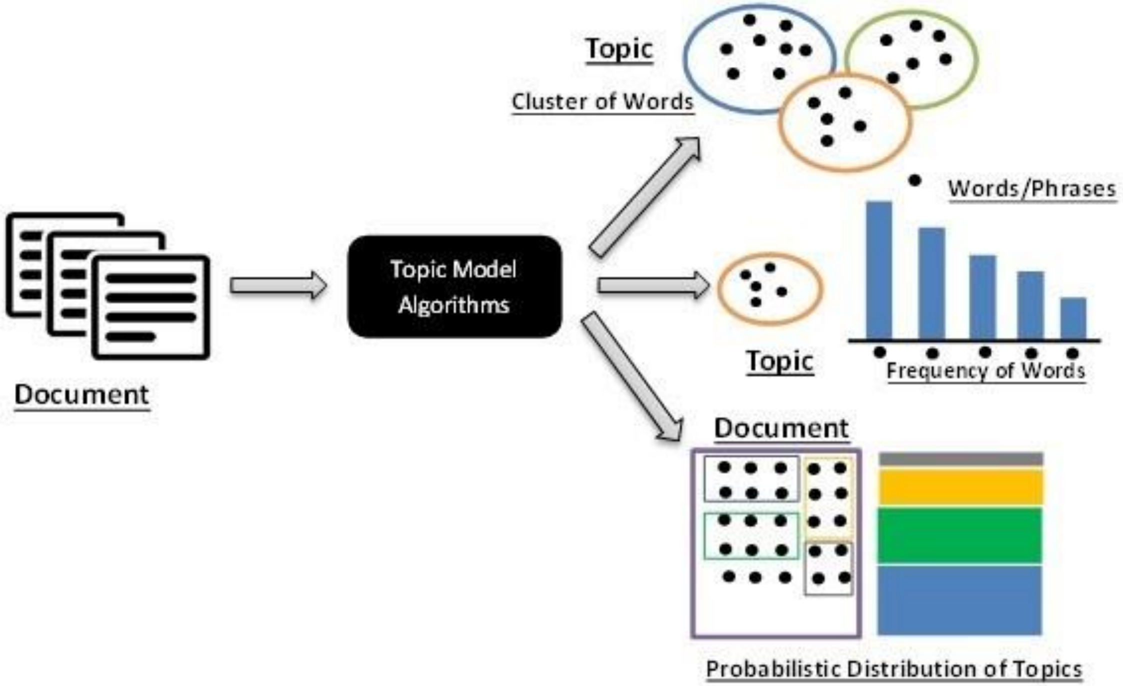


Figure-1

The rapid rise of photorealistic face generation

Representative outputs from original papers and official project pages

Sources: supplied 2014–18 comparison; Karras et al. (StyleGAN, StyleGAN2, StyleGAN3 / NVIDIA).

From plausible pixels to controlled visual generation

Quality gains now include composition, typography, prompt following, editing, and multimodal control

2022
Imagen

2023
SDXL

2024
Stable Diffusion 3

2025
GPT-4o image generation

2026
Muse Image / Muse family

Sources: Saharia et al. (Imagen); Podell et al. (SDXL); Esser et al. (SD3); official OpenAI and Meta launch pages.

Generative priors recover images from incomplete measurements

From task-specific generators to general diffusion priors and posterior sampling

EARLIER MILESTONES

Super-resolution — PULSE · 2020

Inpainting — task-specific deep prior

Colorization — learned image statistics

FOUNDATION PRIORS + POSTERIOR SAMPLING

Real-world restoration & super-resolution — SUPIR · 2024

Inpainting — DAPS · 2025

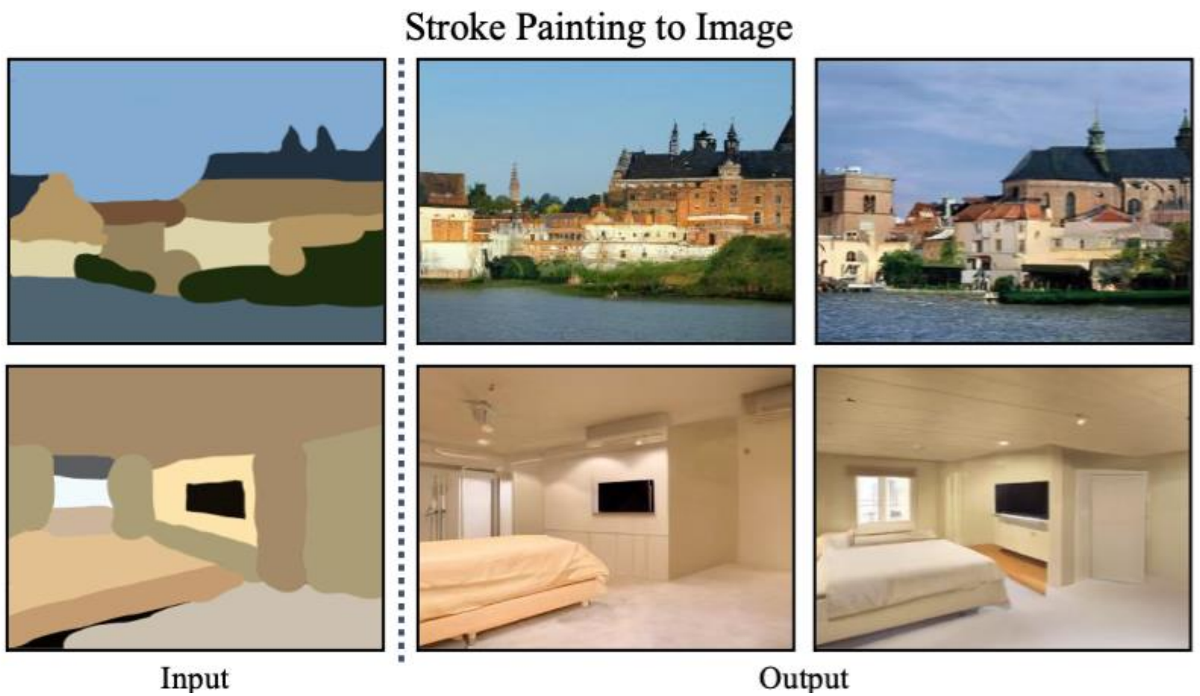
masked measurement → posterior sample

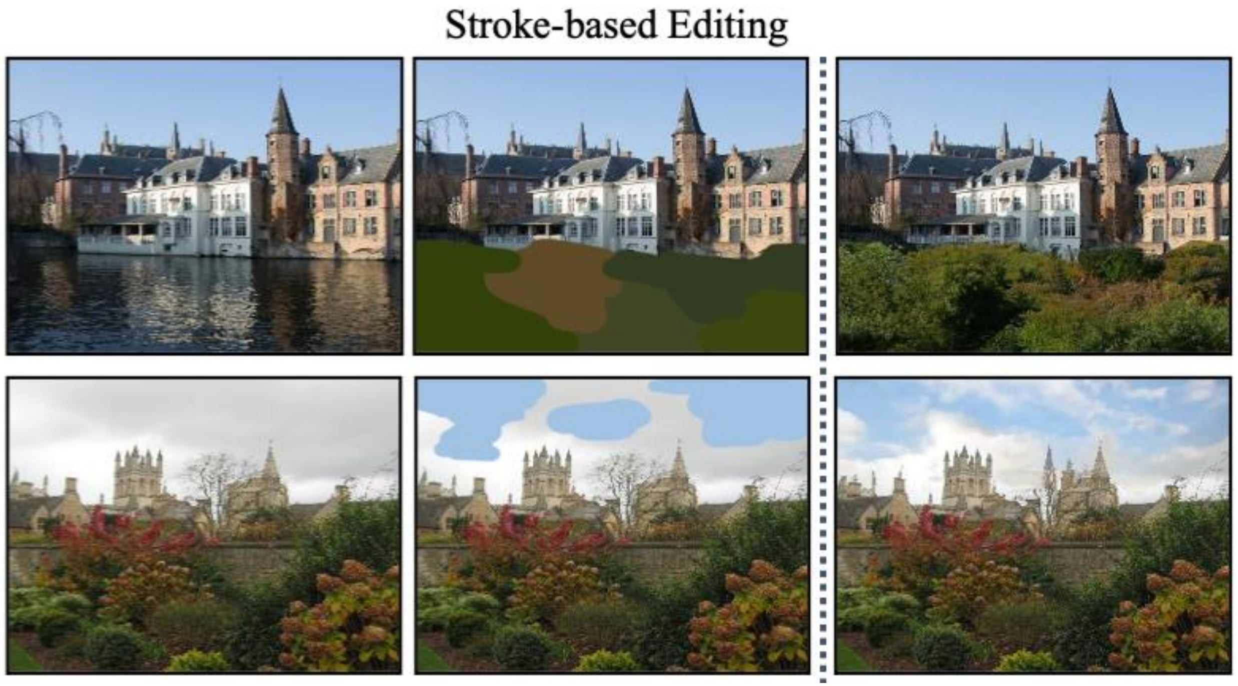
Text-guided colorization — Diffusing Colors · 2023

grayscale measurement → semantically controlled color

Sources: supplied reference; Menon et al. (PULSE); Fanghua Yu et al. (SUPIR); Zhang et al. (DAPS); Frans et al. (Diffusing Colors).

Progress in Inverse Problems





Video Generation

2001 dynamic texture

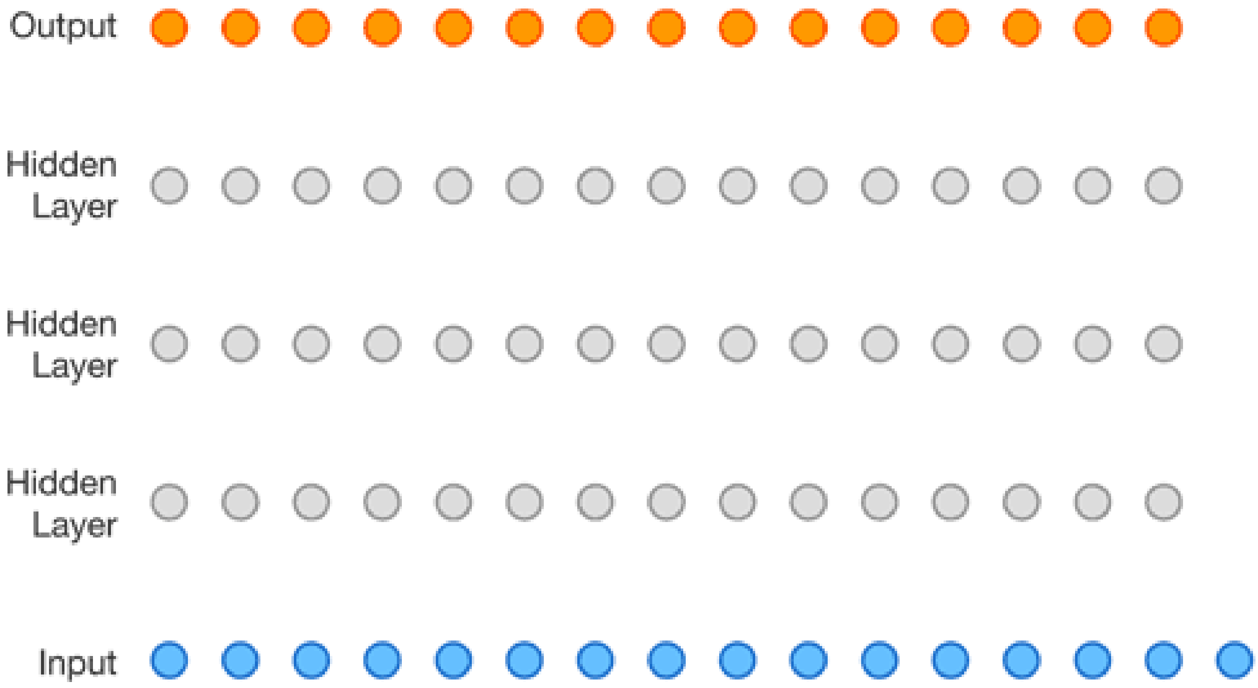
2016 Video GAN

2021 VideoGPT

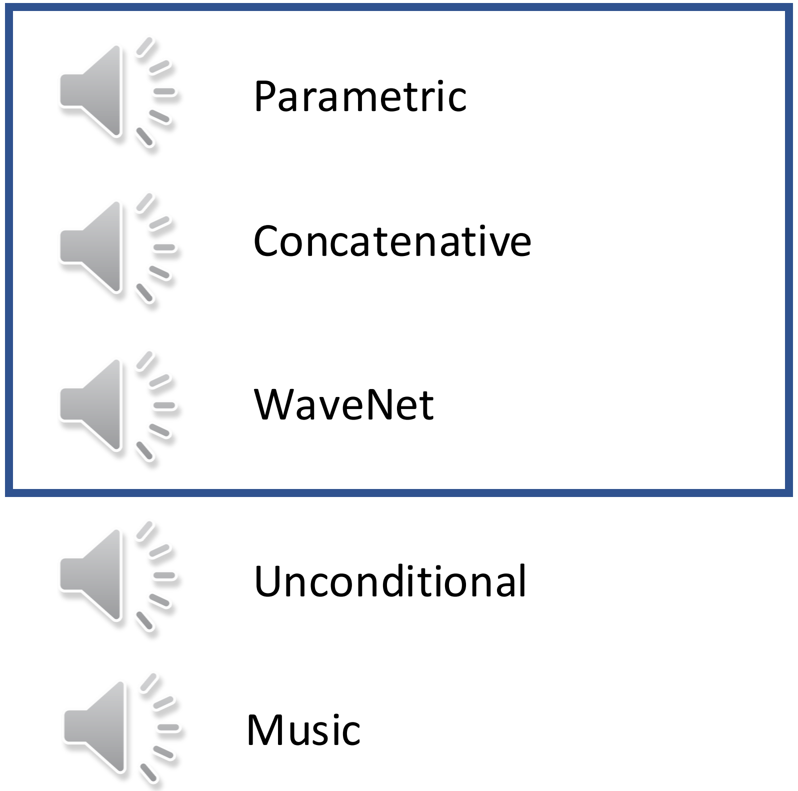
2026 Muse Video

WaveNet

Generative model of speech signals



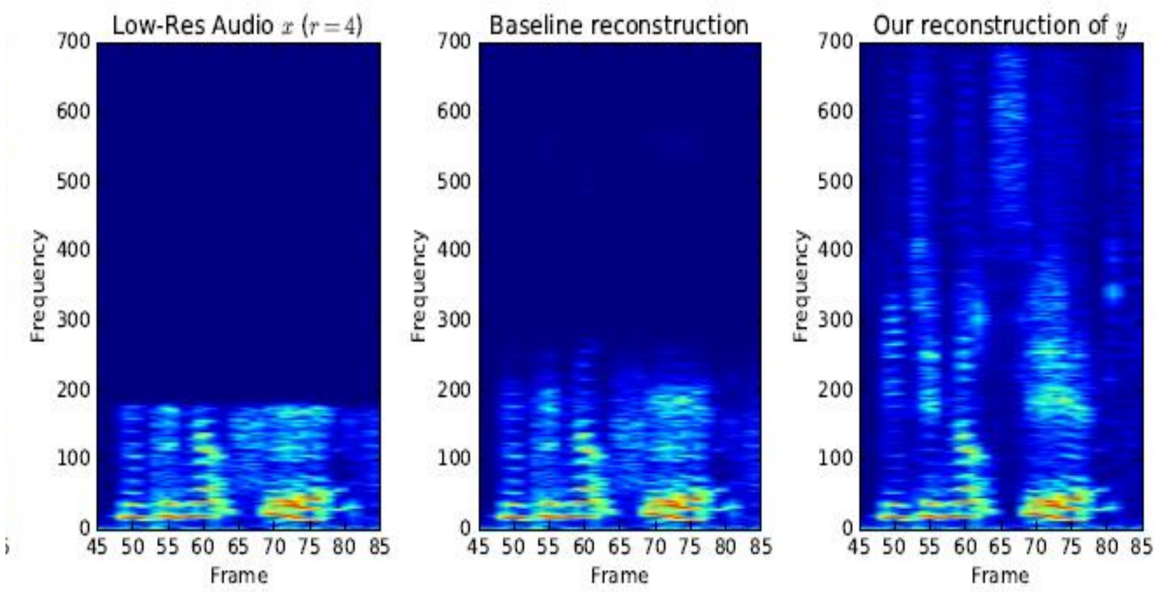
Text to Speech



van den Oord et al, 2016c

Audio Super Resolution

Conditional generative model P(high-res signal | low-res audio signal)



Low res signal

High res audio signal

Kuleshov et al., 2017

Machine Translation

Conditional generative model P(English text| Chinese text)

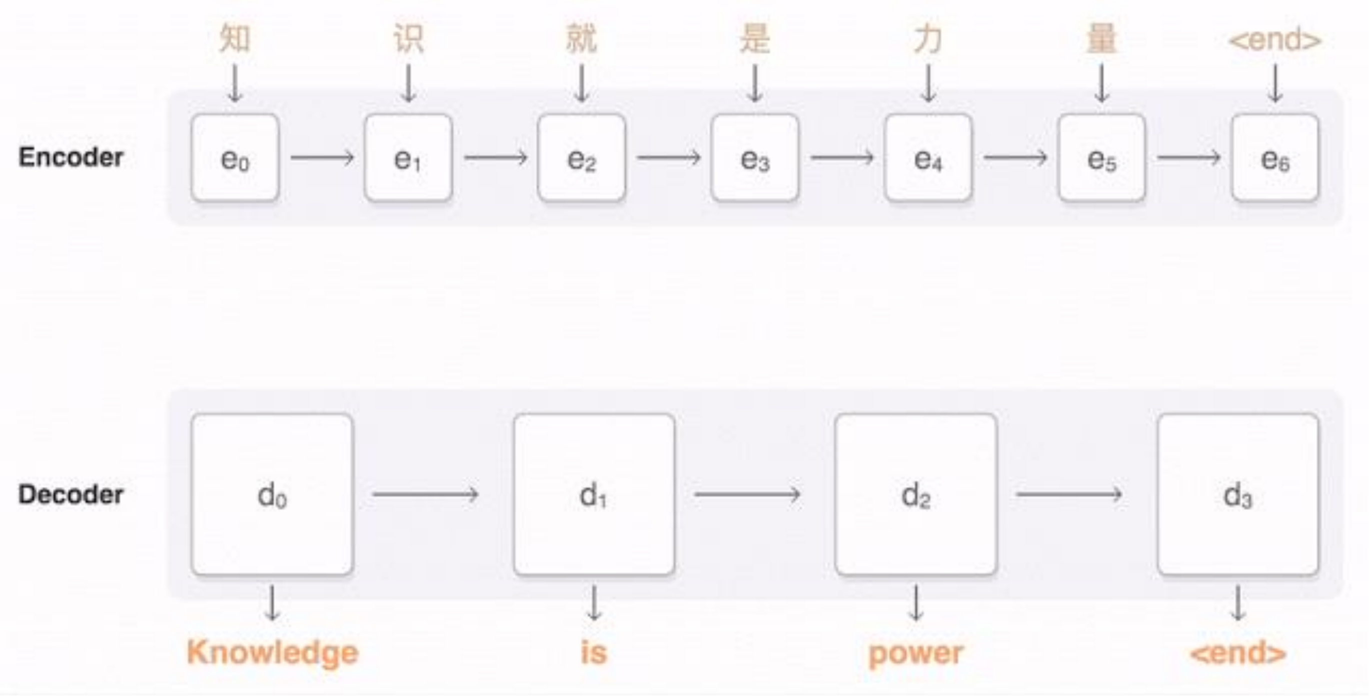


Figure from Google AI research blog.

Language Generation

Custom prompt

To get an A+ in deep generative models, students have to

Completion

To get an A+ in deep generative models, students have to be willing to work with problems that are a whole lot more interesting than, say, the ones that most students work on in class. If you're a great student, the question above can be avoided and you'll be able to do great work, but if you're not, you will need to go beyond the basics before getting good.

Now to be clear, this advice is not just for the deep-learning crowd; it is good advice for any student who is taking his or her first course in machine learning.

The key point is that if you have a deep, deep brain of a computer scientist, that's just as important to you.

P(next word | previous words)

Radford et al., 2019
Demo from talktotransformer.com

Code Generation

parse_expenses.py write_sql.go sentiment.ts addresses.rb

```python
import datetime

def parse_expenses(expenses_string):
    """Parse the list of expenses and return the list of triples (date, va
```

OpenAI Codex

Images and Text

TEXT PROMPT an armchair in the shape of an avocado. . . .

AI-GENERATED IMAGES

Edit prompt or view more images↓

P(image | caption)

TEXT PROMPT a store front that has the word ‘openai’ written on it. . . .

AI-GENERATED IMAGES

DeepFakes

Which image is real?

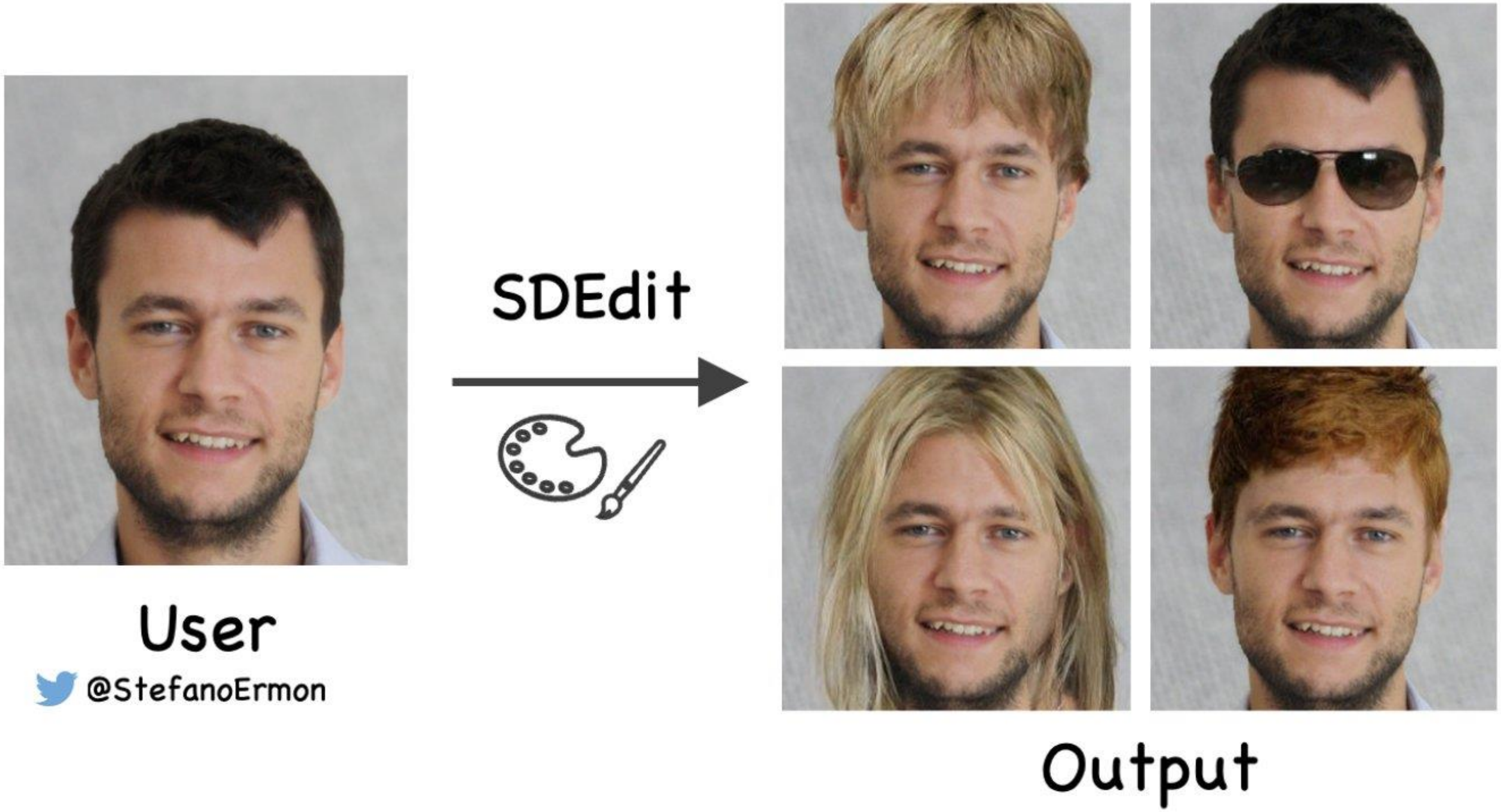


Image Translation

Conditional generative model P(zebra images| horse images)

Zhu et al., 2017